

UEC311: CIRCUIT ANALYSIS AND SYNTHESIS				
	L	T	P	Cr
	2	1	0	2.5
Course Objective: To acquaint the students with the fundamental principles of circuit theory and network synthesis.				
Syllabus Introduction: Review of KCL, KVL, Source transformations, Network theorems– Superposition, Reciprocity, Thevenin, Norton, Maximum power transfer, Tellengen’s theorem. Two Port Network Descriptions: Two-port description in terms of Z-, Y-, ABCD-parameters, Image parameters, Inter-connection of two-port network. Network Functions: Definition of Laplace transform (LT) & its properties, Concept of complex frequency & s-plane, Introduction to basic functions and their LT, Transform impedances, Network functions of one-port. Time Domain Analysis: Solution of networks – RL & RC series using Laplace transform and their Steady state analysis. Network Synthesis: Concept of Poles & Zero, Pole-Zero diagram, Stability, Hurwitz polynomials, Positive real functions (PRF), Synthesis of One-port networks with two kinds of circuit elements, Synthesis of Driving-point functions, basic synthesis procedure, Methods of Synthesis – Foster and Cauer forms.				
Laboratory Work: NA				
Course Learning Objectives (CLO) The students will be able to: 1. Apply various network theorems to circuits. 2. Analyze the two-port networks. 3. Apply Laplace transformation techniques to analyze the transient behavior of the electrical circuits. 4. Determine the stability of network function. 5. Synthesize one port electrical network.				
Text Books 1. D Roy Choudhary, <i>Network Analysis & Synthesis</i>, NewAge International, (2013), 2nd ed. 2. P. Ramesh Babu, <i>Electric Circuit Analysis</i>, SCITECH, (2018), 2nd ed. 3. Sudhakar, A., <i>Circuits and Networks</i>, Tata McGraw Hill (2010) 4th ed.				
Reference Books 1. Vanvalkenberg, M.E., <i>Networks Analysis</i>, Prentice Hall of India (2007) 3rd ed. 2. Kuo, F., <i>Network Analysis and Synthesis</i>, John Wiley (2003) 2nd ed. 3. Anderson, B.D.O., Vongpanitlerd, S., <i>Network Analysis and Synthesis</i>, Dover Publications (2006) 3rd ed.				